VANISHING OF SIMPLICIAL COHOMOLOGY FROM COMPONENTWISE VANISHING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stub remark-simplicial-cohomology-site; external tool from More on Algebra.

Editorial note. The label lemma-simplicial-sheaf-cohomology-site-zero names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Simplicial cohomology double complex. Same unit, Remark remark-simplicial-cohomology-site. RGamma(C_total, K) is represented by the totalization Tot_pi of the double complex A^{p,q}.
- Totalization vanishing. More on Algebra, Lemma more-algebra-lemma-vanishing-coh-prod-totalization. If H^{-p}(A^{p, bullet}) = 0 for all p then the cohomology of the product totalization vanishes in degree 0.

Source reference. Original lemma 0H0Q: sources/stacks-chow/tags.txt, source lines 21220–21220. Prerequisite pointer invoked at proof line(s) 1878; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0H0V: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1813–1813. Prerequisite pointer invoked at proof line(s) 1873; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 1862; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE. Let K be an object of $D(\mathcal{C}_{total})$.*

(1) *If $H^{-p}(\mathcal{C}_p, K_p) = 0$ for all $p \geq 0$, then $H^0(\mathcal{C}_{total}, K) = 0$.*
(2) *If $R\Gamma(\mathcal{C}_p, K_p) = 0$ for all $p \geq 0$, then $R\Gamma(\mathcal{C}_{total}, K) = 0$.*

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Proof. With notation as in Remark 0H0V we see that $R\Gamma(\mathcal{C}_{total}, K)$ is represented by $\mathrm{Tot}_\pi(A^{\bullet,\bullet})$. The assumption in (1) tells us that $H^{-p}(A^{p,\bullet}) = 0$. Thus the vanishing in (1) follows from More on Algebra, Lemma 0H0Q. Part (2) follows from part (1) and taking shifts. $\square$